

G. D. M. Goel

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Education

- 2030 **(Expected) Ph.D., Mathematics**, Princeton University, Princeton, NJ
- 2026 **M.A. Mathematics**, Princeton University, Princeton, NJ
- 2025 **MASt in Mathematics (Pure Mathematics)**, University of Cambridge, Cambridge, UK
Part III of the Mathematics Tripos at Trinity College (*pass with distinction with distinguished performance*)
- 2024 **A.B.**, Harvard University, Cambridge, MA
Double concentration in Mathematics & Germanic Languages and Literatures with secondary in Music
(GPA 4.0, *summa cum laude*, *highest honors in mathematics*)
- 2019 *Pace Junior Science College Andheri (PJSC)*, Mumbai, India
- 2017 *Thakur Public School (TPS)*, Mumbai, India

Writings

Publications and Preprints

- [1] The Chow Ring Classes of PGL_3 Orbit Closures in $\mathbb{G}(1, 5)$. 2022. [\[arXiv Preprint\]](#)
Presented at the HCURA [National Collegiate Research Conference \(NCRC\)](#) 2023.
- [2] On Powerful Integers Expressible as Sums of Two Coprime Fourth Powers (with [Noam D. Elkies](#)).
Res. Number Theory **9**, 78 (2023) (proceedings of ANTS-XV, 2022). doi: [10.1007/s40993-022-00415-9](https://doi.org/10.1007/s40993-022-00415-9)

Research Article(s) In Preparation

- [3] Chow Classes of the Varieties of Secant and Tangent Lines to a Projective Variety. Advisor: [Hannah Larson](#).
- Theses**
- [4] Galois Cohomology (and the Isogeny Invariance of the Strong Birch and Swinnerton-Dyer Conjecture for Abelian Varieties). Part III Essay. Advisor: [Jack Thorne](#).
- [5] The Joys of the Atiyah-Singer Index Theorem. Undergraduate Senior Thesis. Advisors: [Joe Harris](#) and [Dan Freed](#). Recipient of the Hoopes Prize, the Fay Prize, and the Friends Prize.

These and other expository articles are available on [my webpage](#).

Teaching and Mentorship

- 2025 - 26 **Mentor, Directed Reading Program, Princeton**: supervised reading projects on complex algebraic geometry (S26), representation theory (S26), Riemann surfaces (F25), and algebraic topology (F25).

Ross Mathematics Program USA

- Summer 2024 **Visiting Lecturer** at Ross/Ohio.

Course Head, *Plane Algebraic Curves*. Designed and taught this course on classical algebraic geometry aimed at returning students and counselors. In addition, gave two other mini lecture series for counselors:

Mathematical Ghostbusters: Despookifying Spectral Sequences (see [lecture notes in progress](#)) and **Cohomology of Curves** (see [lecture notes](#)).

- Summer 2020 **Counselor** at Ross/Zoom. Led discussion sessions for high school students for a number theory course.

Course Assistant, *Rational Points on Curves*, an advanced course for counselors taught by [Stefan Patrikis](#).

Course Assistant, Department of Mathematics, Harvard University

Led discussion and review sessions, held office hours, graded homework, and prepared solutions.

Spring 2024 **Math 222: Lie Groups and Lie Algebras**, taught by [Noam D. Elkies](#); see the [course webpage](#).

Fall 2023 **Math 213A: Advanced Complex Analysis**, taught by [Curt McMullen](#); see the [course webpage](#).

Spring 2023 **Math 129: Number Fields**, taught by [Salim Tayou](#).

Fall 2022 **Math 114: Analysis of Function Spaces, Measure and Integration**, taught by [H. T. Yau](#).

Spring 2022 **Math 132: Differential Topology**, taught by [Dennis Gaitsgory](#).

Fall 2021 **Math 55A: Honors Group Theory and Abstract Algebra**, taught by [Denis Auroux](#).

In addition, served as a volunteer course assistant for the following.

Spring 2021 **Math 132**, taught by [Joe Harris](#).

Spring 2021 **Math 55B: Honors Real and Complex Analysis**, taught by [Denis Auroux](#).

Fall 2021 **Math 55A**, taught by [Denis Auroux](#).

Other

Summer 2020 **Course Head, Math 57, Math Online Summer Program**

Taught group theory to high school students as part of a program organized by [Alec Sun](#).

2018 - 19 **Teaching Assistant, Math Olympiad Program, PJSC**

Taught students in grades 8 through 12 advanced math in preparation for the IMO.

Selected Talks Given

Invited

April 2024 **The Atiyah-Singer Index Theorem and Almost Complex Spheres**,
Special talk by the Friends Prize Recipient, Harvard University

April 2024 **The Chow Ring Classes of PGL_3 Orbit Closures in $\mathbb{G}(1,5)$** , gradieNTAG, Simon Fraser University

March 2024 **Chow Classes of Varieties of Secant and Tangent Lines**,
Special Student Algebraic Geometry Seminar, Stanford University

Contributed

Dec 2025 **The Weil Conjectures**, Princeton First-Year Seminar

Nov 2025 **Galois Theory Like You've Never Seen Before**, Princeton Graduate Student Seminar

May 2025 **Bertrand's Postulate (or Why All of Maths is Cool)**, The Trinity Maths Society Easter Peer Talks

March 2025 **An Introduction to Moduli Spaces in Algebraic Geometry**, Part III Seminar Series, Lent 2025

July 2024 **Indra's Pearls: Symmetry, Recursion, and Beauty in Mathematics**, Ross/Ohio

June 2024 **What is Sharkovsky's Theorem, or why does period three imply chaos?**, What is ... ? Seminar, OSU

June 2024 **Can You Solve It (It = A Quintic)? Topology Says No**, Ross/Ohio

May 2024 **(Almost) Complex Structures on 4-Manifolds**, Harvard University Math Table

Nov 2022 **Emmy Noether: Mathematician Extraordinaire**, HMMT Education Session

Aug 2022 **On Powerful Integers Expressible as Sums of Two Coprime Fourth Powers**, ANTS-XV, U. of Bristol

Feb 2022 **The Three-Sphere in Dante's *Divine Comedy***, HMMT Education Session

Nov 2021 **The Inverse [Quadratic] Central Potential and the Cubic Equation**
Harvard Society of Physics Students (SPS), Undergraduate Chalk Talks Series

Aug 2021 **Galois Cohomology and Hilbert's Theorem 90**, Ross 2021, Dorm Lecture for *Rational Points on Curves*

Nov 2020 **What is Algebraic Geometry and Why Do We Care?**, HMMT Education Session

Oct 2019 **Fermat's Little Theorem and its Applications to Cryptography**, Harvard University Math Table

A list of all talks given, along with some accompanying notes, can be found on [my webpage](#).

Honors, Awards, and Research Funding

Academic

- 2025 **Tripes Prize**, “to students who have most distinguished themselves in the MAST in Mathematics.”
- 2025 **Examination Prize**, Trinity College, Cambridge
- 2025 **Senior Scholarship**, Trinity College, Cambridge
- 2024 **Lt. Charles H. Fiske III Scholarship**, Harvard-Cambridge Scholarship to Trinity College, Cambridge
- 2024 **Sophia Freund Prize**, “awarded annually to the highest ranking undergraduate(s).”
- 2024 **David B. Mumford Mathematics Prize**, “to the most promising senior concentrator(s) in mathematics.”
- 2023 **Phi Beta Kappa (PBK), Senior 48**, membership in national academic honor society.
- 2023, '22, '20 **John Harvard Scholarship**, awarded to students in the top 5% of their class.

For Writings

- 2024 **Captain Jonathan Fay Prize**, for [5], given to “top theses of Harvard College’s graduating class”.
- 2024 **Thomas Temple Hoopes Prize**, for [5], awarded for “outstanding scholarly work or research.”
- 2024 **Friends Prize**, for the top two theses presented to The Friends of the Harvard Math Department, interviewed in the [2024 Harvard Math Newsletter](#).

Research Funding

- 2022 **Fellow, Herchel Smith Undergraduate Science Research Program**, for [1].
- 2022 **Fellow, Harvard College Program for Research in Science and Engineering (PRISE)**
- 2020 **Harvard College Research Program**, for [2].

Other

- 2024 **Caroline Isenberg Award**, Lowell House, Harvard
- 2023 **Elliot & Mary Perkins Prize**, Lowell House, Harvard
- 2019, '18 **Merit Award, Indian National Mathematics Olympiad (INMO)**
- 2017 **Thakur Ramnarayan Memorial Award, TPS**

Extracurricular Activities

- 2026 - **Co-founder, Princeton $\text{Spec}(\mathbb{Q}_p)$** : Founded and ran the queer math student organization at Princeton.
- 2024 **Mentor, HARMONY, Harvard PBHA**: gave piano lessons to middle school students in Cambridge.
- 2022 - 24 **Member, Harvard MIHNUET**: performed music in hospices and senior living communities.
- Harvard Undegraduate Gender Inclusivity in Mathematics (HUGIIM)**
- 2022 - 23 **Co-president**: Organized projects and socials, collaborated with other inclusivity organizations, and served as student representative in the Math Department Community Committee.
- 2021 - 22 **Mentorships Co-Chair**: Co-organized the *Math Includes* mentorship program.

Arts

- 2021 - 23 **Student Composer, Harvard Choruses New Music Initiative**
- Feb 2022 **Bard, If A Picture Never Lies, Harvard Classics Club**. See [this article in the Harvard Crimson](#).

Choirs

Princeton University Chamber Choir ('25-); [Early Music Princeton Singers](#) ('25-); [Cambridge University Schola Cantorum](#) ('24-25); [Emmanuel College Chapel Choir](#) ('24-25); [Harvard University Choir](#) ('22-24) as a [Ferris Choral Fellow](#) ('23-24); [Harvard Glee Club](#) ('19-24) and [Glee Club Lite](#) ('21-22).

Other

Software: Python, Java, C++, SageMath, Pari/GP, Macaulay2
Languages: English (native), Hindi (native), German (C1), Spanish (B1), French (A1)